# THE GYSIN CLASS AT INFINITY COMMUTES WITH EVERY BIVARIANT CLASS PULLED BACK FROM A LOCALLY FINITE TYPE MORPHISM

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

*Editorial note.* The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

*Editorial note.* Canonical location of the counted proposition: Stacks tag 0FAS (sources/stacks-chow/pinned-original-source.tex line 10156).

*Allowed standard prerequisites (editorial).*

- The Gysin-at-infinity bivariant class and its defining property. Same chapter, Lemma labelled lemma-gysin-at-infty (source line 10026), invoked in the statement. Supplies the class C in A^0(W_infinity -> X) with C cap alpha = i_infinity^* beta, the input to both sides of the commutativity identity.
- Bivariant classes and their compatibility with base change and with Gysin maps. Same chapter, chow.tex, the definition of A^*(Y -> X) and the bivariant-class axioms used in the displayed identity at proof line 10183. Supplies that a bivariant class commutes with the Gysin homomorphism i_{Y,infinity}^* and that the construction is stable under base change; the diagram chase itself is inside the counted span.

*Source reference.* Original lemma 0F9J: sources/stacks-chow/pinned-original-source.tex, source lines 10026–10026. Same-chapter pointer into the author environment lemma at source line 10026; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

**Lemma 0.1.** *In Lemma 0F9J let $f : Y \to X$ be a morphism locally of finite type and $c \in A^*(Y \to X)$. Then $C \circ c = c \circ C$ in $A^*(W_\infty \times_X Y \to X)$.*

*Date*: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Proof.* Consider the commutative diagram

$$\begin{array}{ccccccc} W_\infty \times_X Y & = & W_{Y,\infty} & \xrightarrow[i_{Y,\infty}]{} & W_Y & \xrightarrow[b_Y]{} & \mathbf{P}^1_Y \xrightarrow{p_Y} Y \\ & & \downarrow & & \downarrow & & \downarrow \quad \downarrow f \\ & & W_\infty & \xrightarrow{i_\infty} & W & \xrightarrow{b} & \mathbf{P}^1_X \xrightarrow{p} X \end{array}$$

with cartesian squares. For an elemnent $\alpha \in \mathrm{CH}_k(X)$ choose $\beta \in \mathrm{CH}_{k+1}(W)$ whose restriction to $b^{-1}(\mathbf{A}^1_X)$ is the flat pullback of $\alpha$. Then $c \cap \beta$ is a class in $\mathrm{CH}_*(W_Y)$ whose restriction to $b_Y^{-1}(\mathbf{A}^1_Y)$ is the flat pullback of $c \cap \alpha$. Next, we have

$$i^*_{Y,\infty}(c \cap \beta) = c \cap i^*_\infty \beta$$

because $c$ is a bivariant class. This exactly says that $C \cap c \cap \alpha = c \cap C \cap \alpha$. The same argument works after any base change by $X' \to X$ locally of finite type. This proves the lemma. $\square$